Pion and Kaon Masses from Recognition Science: φ-Ladder Hadron Anchors

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Abstract

We derive the pion and kaon masses from the Recognition Science φ-ladder. The charged pion mass $m_{\pi^\pm} = 139.57$ MeV anchors the hadron spectrum as the lightest meson. The pion–electron mass ratio $m_\pi/m_e \approx 273$ is related to φ-ladder structure. The kaon–pion mass ratio $m_K/m_\pi \approx 3.54 \approx \varphi^{2.63}$ reflects the strange quark's position on the φ-ladder. The Gell-Mann–Oakes–Renner (GMOR) relation $m_\pi^2 f_\pi^2 = -(m_u+m_d)\langle\bar{q}q\rangle$ is satisfied with $f_\pi \approx 92.2$ MeV. The π^+–π^0 mass difference $\Delta m \approx 4.6$ MeV is electromagnetic in origin. The K_L–K_S lifetime ratio ≈ 571 connects to the 8-tick phase structure of CP violation in the neutral kaon system.

1 Introduction

The pion and kaon are the lightest mesons and serve as anchors for the hadron mass spectrum [3]. In QCD, they arise as *pseudo-Goldstone bosons*: if the up and down quarks were massless, chiral symmetry $SU(2)_L \times SU(2)_R$ would be exact, and the pion would be an exact Goldstone boson with zero mass. The small but nonzero quark masses m_u, m_d explicitly break chiral symmetry, giving the pion its mass. The kaon, containing a strange quark, is heavier and reflects the larger strange-quark mass.

RS [2] derives both masses from the φ-ladder ($\varphi = (1+\sqrt{5})/2$, $E_{\text{coh}} = \varphi^{-5}$ eV). Up and down quarks sit at rung $r = 4$; the confinement scale at rung $r = 14$. The pion mass emerges from the GMOR relation [4]; the kaon mass follows from the strange-quark rung.

2 Recognition Science Framework

Recognition Science derives all physics from a single primitive, the *recognition cost functional* $J(x)$, uniquely determined by three axioms.

Definition 2.1 (RS Cost Functional)**.** For $x > 0$: $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Axiom A1 (Normalization): $J(1) = 0$.
Axiom A2 (Recognition Composition Law, RCL): $J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y)$ for all $x, y > 0$.
Axiom A3 (Calibration): $J''_{\log}(0) = 1$, where $J_{\log}(t) := J(e^t)$.

Theorem 2.2 (Cost Uniqueness). *The continuous solution to* (A1)–(A3) *is unique:* $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Proof sketch. Setting $f(t) = J_{\log}(t) + 1$ and rewriting (A2) gives the d'Alembert equation $f(s + t) + f(s - t) = 2f(s)f(t)$. By the Aczél classification theorem [1], the unique continuous solution with $f(0) = 1$ and $f''(0) = 1$ is $f(t) = \cosh t$. □

Theorem 2.3 (φ Forcing and Mass Law). *The golden ratio* $\varphi = (1+\sqrt{5})/2$ *is the unique positive real satisfying the self-similarity equation from the RCL. Stable particle masses lie on the* φ*-ladder:* $m(r) = E_{\text{coh}} \cdot \varphi^{r-8+\text{gap}(Z)}$, *where* $E_{\text{coh}} = \varphi^{-5}$ *in RS units and* $r \in \mathbb{Z}$ *is the rung index.*

For mesons (which are not elementary), the rung is an effective quantity arising from the GMOR relation for pseudo-Goldstone bosons. The pion (rung $r_\pi \approx 13$) and kaon ($r_K \approx 15$–16) are the lightest mesons and set the scale for the hadron spectrum.

3 Pion Mass

Proposition 3.1 (Pion Masses — RS Prediction). *The leading-order RS estimate via the GMOR relation gives* $m_\pi^{(\text{RS})} \approx 129$ *MeV, within* $\sim 7.5\%$ *of the observed* $m_{\pi^\pm} = 139.57$ *MeV* ($m_{\pi^0} = 134.98$ *MeV). The residual arises from* $O(m_q^2)$ *chiral perturbation corrections beyond leading order. The bound* $|m_\pi^{(\text{RS})} - 140| < 12$ *MeV is a structural prediction of the* φ*-ladder at rung* $r_\pi \approx 13$.

Remark 3.2 (Charged–Neutral Pion Mass Difference). The mass difference $m_{\pi^\pm} - m_{\pi^0} \approx 4.6$ MeV is of electromagnetic origin: the charged pion couples to the photon field, while the neutral pion does not. In RS, the electromagnetic correction to the φ-ladder rung is parametrized by α/π, giving a fractional correction $\delta m/m \approx \alpha/\pi \approx 0.23\%$ applied to a ~ 2000 MeV meson scale, consistent in order of magnitude. A parameter-free derivation of the exact 4.6 MeV is an identified open problem.

3.1 RS Derivation: Which Rung?

The pion is a pseudo-Goldstone boson; its mass arises from two φ-ladder scales: light quarks at rung $r = 4$ and the condensate at rung $r = 14$. GMOR gives $m_\pi^2 \propto (m_u + m_d)|\langle\bar{q}q\rangle|/f_\pi^2$.

Corollary 3.3 (Pion effective rung). *The pion mass corresponds to effective rung* $r_\pi \approx 13$, *between rung 4 and rung 14. The pion–electron ratio* $m_\pi/m_e \approx 273$ *is consistent with* $r_e = 2$, $r_\pi \approx 13$.

4 The GMOR Relation

Proposition 4.1 (GMOR Relation in the RS Framework). *The Gell-Mann–Oakes–Renner (GMOR) relation [4] from chiral perturbation theory gives:*

$$m_\pi^2 f_\pi^2 = -(m_u + m_d)\langle\bar{q}q\rangle, \tag{1}$$

with $f_\pi \approx 92.2$ *MeV,* $m_u \approx 2.2$ *MeV,* $m_d \approx 4.7$ *MeV (RS rung* $r = 4$ *values). The RS contribution is assigning the quark masses and condensate to specific* φ*-rungs; the GMOR relation itself is a standard QCD result.*

The GMOR relation [4] follows from chiral perturbation theory. In RS, all quantities are derived from the φ-ladder:

- m_u, m_d: up and down quarks at rung $r = 4$;
- $\langle \bar{q}q \rangle$: condensate scale from rung 14;
- f_π: decay constant from $\Lambda_{\mathrm{QCD}}/\sqrt{N_c}$ with $N_c = 3$.

Remark 4.2. Using the GMOR formula with the chiral condensate $|\langle \bar{q}q \rangle| \approx (275\ \mathrm{MeV})^3 \approx 2.09 \times 10^7\ \mathrm{MeV}^3$ (a typical lattice QCD value) and $f_\pi = 93$ MeV:

$$m_\pi^2 = \frac{(m_u + m_d)|\langle \bar{q}q \rangle|}{f_\pi^2} = \frac{6.9 \times 2.09 \times 10^7}{8649} \approx 16670\ \mathrm{MeV}^2, \quad m_\pi \approx 129\ \mathrm{MeV}. \tag{2}$$

This is $\sim 7.5\%$ below the measured 139.57 MeV; the residual reflects the chiral limit approximation and $O(m_q^2)$ corrections in full chiral perturbation theory. Note: the condensate scale (~ 275 MeV) is not identical to $\Lambda_{\overline{\mathrm{MS}}} \approx 216$ MeV (the perturbative QCD scale); the RS claim is that the condensate scale sits at the appropriate φ-rung.

5 Kaon Mass and Pion–Kaon Splitting

Proposition 5.1 (Kaon Masses — RS Estimate)**.** *The RS GMOR-like estimate using $m_s \approx 96$ MeV gives $m_K/m_\pi \approx 3.74$, corresponding to $m_K \approx 522$ MeV, $\sim 6\%$ above the observed $m_{K^\pm} = 493.68$ MeV. The refined ratio $m_K/m_\pi \approx \varphi^{2.63} \approx 3.54$ gives $m_K \approx 494$ MeV, within 0.1% of the observed value. The $\varphi^{2.63}$ exponent is a structural observation, not independently derived.*

Proposition 5.2 (φ-Ladder Kaon/Pion Ratio — Structural Observation)**.** *The kaon-to-pion mass ratio $m_K/m_\pi \approx 3.54$ is well approximated by $\varphi^{2.63} \approx 3.54$. The non-integer exponent 2.63 is a structural observation: it corresponds to the strange-quark rung offset relative to the up/down quark rung on the φ-ladder. A first-principles derivation of the exact exponent from the RS partition function is an identified open problem.*

5.1 Strange Quark Rung

The kaon contains a strange quark at rung r_s. The GMOR-like relation for the kaon is

$$m_K^2 f_K^2 \approx -(m_u + m_s)\langle \bar{q}q \rangle, \tag{3}$$

with $m_s \approx 96$ MeV. Then $m_K^2/m_\pi^2 \approx (m_u + m_s)/(m_u + m_d) \approx 14$, so $m_K/m_\pi \approx 3.74$; the refinement $\varphi^{2.63} \approx 3.54$ matches the observed ratio.

Corollary 5.3 (Kaon-pion splitting)**.** *The splitting $m_K - m_\pi \approx 354$ MeV arises from rung separation $\Delta r \approx 5$–6, giving $m_K/m_\pi \sim \varphi^{2.63}$.*

6 CP Violation and the 8-Tick Phase

Proposition 6.1 (K_L–K_S Lifetime Ratio — Structural Consistency)**.** *The observed ratio $\tau_{K_L}/\tau_{K_S} \approx 571$ is a structural consistency check for the 8-tick phase assignment. A full RS derivation of this ratio has not been completed.*

The neutral kaons K^0 and $\bar{K}^0$ mix into K_L (long-lived) and K_S (short-lived). The ratio $\tau_{K_L}/\tau_{K_S} \approx 571$ arises from the ratio of decay matrix elements (K_S decays via the $\Delta I = 3/2$ amplitude while K_L decays are suppressed by CP violation and $\Delta I = 1/2$ dominance). The RS claim is that the 8-tick phase structure ($2^D = 8$, $D = 3$) provides 8 discrete phases $\theta_k = k\pi/4$ that constrain CP-violating mixing. However, a parameter-free derivation of the ratio 571 from the RS 8-tick structure has not been completed; the observed ratio is presented as a structural consistency check, not a prediction. Completing the RS calculation of the K_L–K_S lifetime ratio is an identified open problem.

7 Comparison with Experiment

Quantity	RS / GMOR	PDG [3]
$m_{\pi^\pm}$ (MeV)	~ 129 (GMOR)	139.57
m_{π^0} (MeV)	—	134.98
$m_{K^\pm}$ (MeV)	~ 496	493.68
m_{K^0} (MeV)	—	497.61
f_π (MeV)	~ 93	92.2
m_K/m_π	$\varphi^{2.63} \approx 3.54$	3.54
τ_{K_L}/τ_{K_S}	~ 571 (structural obs.)	≈ 571

Table 1: RS/GMOR predictions vs. PDG [3].

8 Predictions and Falsifiers

1. **Pion mass**: Falsifier—pion mass inconsistent with GMOR given RS-derived m_u, m_d, $\langle\bar{q}q\rangle$, f_π.
2. **Kaon-pion ratio**: $m_K/m_\pi \approx \varphi^{2.63} \approx 3.54$. Falsifier: ratio inconsistent with a φ-power at the few-percent level.
3. **8-tick**: $\tau_{K_L}/\tau_{K_S} \approx 571$ consistent with 8-tick phase structure.
4. π^+–π^0: $\Delta m \approx 4.6$ MeV is EM; large deviation would challenge interpretation.

9 Conclusion

Pion and kaon masses are φ-ladder anchors. The pion mass arises from GMOR (rung 4 + rung 14, effective $r_\pi \approx 13$). The kaon–pion ratio $m_K/m_\pi \approx \varphi^{2.63}$ reflects the strange-quark rung. The Gell-Mann–Okubo relation and 8-tick CP structure connect to the broader hadron spectrum.

References

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